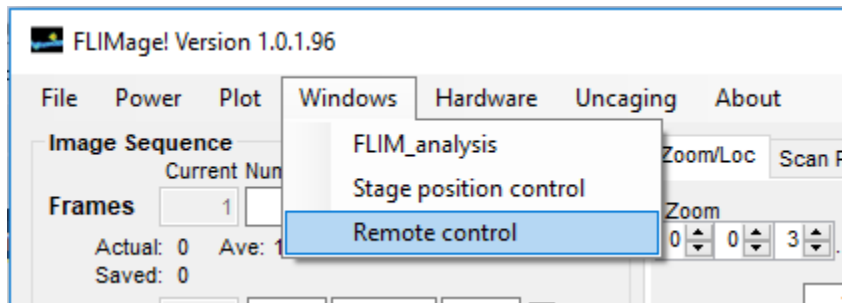
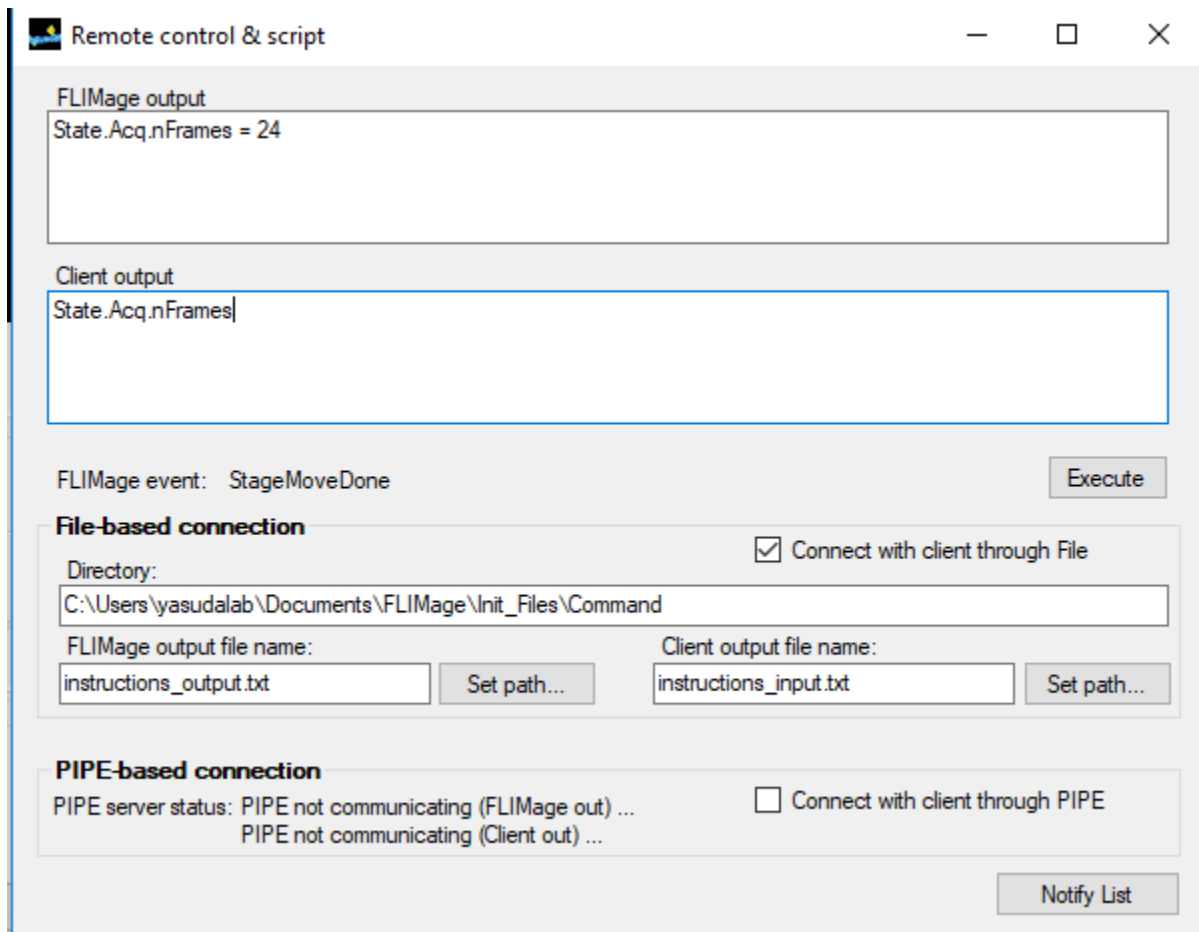


Remote control can be done either through text file or PIPE.

For controlling this feature, click “Windows menu” and then “Remote control” (see below).



You will see a window named “Remote control & script” pops up (see below).



Text file based communication is much easier than PIPE-based communication. You simply use files for output and input to communicate. For now, both output and input instructions need to be in the same folder. Press “set path” to select a folder and files for communication. The line must be appended to the last line of the file created. In C#, it will be something like:

```
File.AppendAllText(FilePath, Command + "\r\n");
```

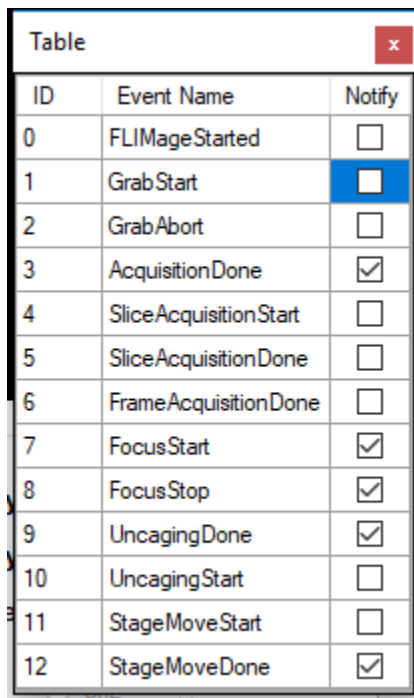
For opening of command file, make sure to use read-only mode.

If you want to use PIPE, turn on “Connect with client through PIPE”, and then turn on client PIPE. Handshaking and so on will be done through PIPE server. An example script for will be found in the folder (FLIM_pipeClient.py). For now, PIPE name is set to FLIMageW (for client to write) and FLIMageR (for client to read). Immediately after establishing connection, FLIMage will send a code “FLIMage”. Client needs to return “FLIMage” to make sure the connection. PIPE will be 1 command – to 1 response communication.

One can simulate sending commands from client by typing commands in “Client output” window, and press “Execute”.

Event notification from FLIMage to Client

To view the event notifications from FLIMage, press "Notify List" button at the bottom in the "Remote control & script" window:



ID	Event Name	Notify
0	FLIMageStarted	☐
1	GrabStart	☒
2	GrabAbort	☐
3	AcquisitionDone	☒
4	SliceAcquisitionStart	☐
5	SliceAcquisitionDone	☐
6	FrameAcquisitionDone	☐
7	FocusStart	☒
8	FocusStop	☒
9	UncagingDone	☒
10	UncagingStart	☐
11	StageMoveStart	☐
12	StageMoveDone	☒

-- sent after FLIMage start ---
command: FLIMageStarted
arguments: none

-- set after Grab start --
command: GrabStart

arguments: none

--sent after movement to new coordinates is complete--

command = StageMoveDone

arguments: x, y, z

example: StageMoveDone,1,34.2,-2

--sent after slice acquisition is done

Command = SliceAcquisitionDone

arguments; {current slice}, {total number of slices}

Execution command from Client to FLIMage

In general, there will be no return. It will execute command.

--Move to new stage position--

command = SetMotorPosition

arguments: x y z

example: SetMotorPosition,12,89.2,0

--Move Piezo motor one step—

Command = MovePiezoStep

Arguments: micrometer (float)

Example: MovePiezoStep, 1.0

--Start grab --

command = StartGrab

--Set Z center position –

(Assign the current Z position to the center of the Z stack)

command = SetCenter

--Start loop –

(Use parameters in “Loop Sequence” panel)

command = StartLoop

--Stop loop –

command = StopLoop

--Abort grab—

command = AbortGrab

--Start uncaging—

command = StartUncaging

--Stop uncaging--

command = StopUncaging

--Set uncaging position--

command = SetUncagingLocation

arguments: ROI_x_pixels, ROI_y_pixels

example: SetUncagingLocation,37,42

return: none

--Create uncaging position--

command = CreateUncagingLocation

arguments: ROI_x_pixels, ROI_y_pixels

example: CreateUncagingLocation,37,42

return: none

--Clear uncaging position--

command = ClearUncagingLocation

--Start Digital Output --

command = StartDO

--Stop Digital Output --

command = StopDO

Parameter request (set, get) from Client to FLIMage

You can access to all parameters by typing the parameter name (see FLIMage\default.txt):

Example: State.Acq.nFrame

Return exmaple: State.Acq.nFrame = 24

All parameters can be set by inserting values as follows:

Example: State.Acq.nFrame = 24

Return: State.Acq.nFrame = 24

Other parameter requests:

--Load setting file --

Command: LoadSetting

Arguments: file number or file name

Example: LoadSetting, 1

Example2: LoadSetting, C:\Users\yasudalab\Documents\FLIMage\Init_Files\Uncaging_imaging.txt

Example3: LoadSetting, Uncaging_imaging.txt

Return: Setting, file number or file name

Note: If there is it is not a full path, it will try default path (... FLImage\Init_Files\)

--Set Zoom to specific value--

command = SetZoom

arguments: zoom

example: setZoom,20

return: Zoom, {zoom value}

--Request for current position to be sent--

command = GetCurrentPosition

arguments: None

return: CurrentPosition, {x}, {y}, {z}

--Request for FOV size--

command = GetFOVXY

arguments: None

return: FovXYum, {x value}, {y value}

--Request Galvo Scanning Position--

command = GetScanVoltageXY

arguments: None

example: GetScanVoltageXY

return: ScanVoltageXY, {x value}, {y value}

--Set Galvo Scanning Position--

command = SetScanVoltageXY

arguments: x, y (in voltage)

example: SetScanVoltageXY,0.2,-4

return: ScanVoltageXY, {x value}, {y value}

--Request Galvo Scan Voltage Multiplier--

command = GetScanVoltageMultiplier

arguments: None

return: ScanVoltageMultiplier, {x value}, {y value}

--Request Galvo Scan Voltage Range Reference--

command = GetScanVoltageRangeReference

arguments: None

return: ScanVoltageRangeReference, {x value}, {y value}

--Set Number of Z Slices--

command = SetZSliceNum

arguments: zSliceNum

example: SetZSliceNum,10

return: ZSliceNum, {z slice number}

--Set X,Y Imaging Resolution--

command = SetResolutionXY

arguments: x,y

example: SetResolutionXY,128,128

return: ResolutionXY, {x pixel number}, {y pixel number}

--Get X,Y Imaging Resolution--

command = GetResolutionXY

arguments: None

example: GetResolutionXY

return: ResolutionXY, {x pixel number}, {y pixel number}

--Convert pixel into voltage--

command = PixelToVoltage

arguments: xpixel, ypixel

example: PixelToVoltage, 64, 64

return: PixelToVoltage, 0, 0

-- Turn Intensity Image saving On/Off--

Command = SetIntensitySaving

Arguments: 0 or 1 (for on or off)

Return: IntensitySaving, {0 or 1}

--Request Image File Path--

Command = GetIntensityFilePath

Arguments: None

Return: IntensityFilePath, {file path}

--Set channels to be saved--

Command=SetChannelsToBeSaved

Arguments: {channel to be saved}

Return: ChannelsToBeSaved, {channel to be saved}

Note: if channel == -1 (default), all channels will be saved.

--Ask if images are acquired--

Command = IsGrabbing

Return: 0 or 1

--Ask if uncaging is going--

Command = IsUncaging

Return: 0 or 1

--Ask if digital output is going--

Command = IsDORunning

Return 0 or 1

--Get FLIMage version--

Command = GetVersion

Return: version#

--Set DIO output on DIO panel --

Command = SetDIOPanel

Arguments: {port#:0-6}, {bool:0/1} [, {port#:0-6}, {bool:0/1}...]

--"Focus" button--

Command = Focus

Arguments: none

--"Stop" button in Stage panel--

Command = StopMotor

Arguments: none

--"Zero all" button in Stage panel--

Command = SetZeroAll

Arguments: none

--"Analyze after each acquisition" checkbox--

Command = SetAnalyzeAfterAcq

Arguments: {bool: 0/1}

--Save "Setting-#.ini"--

Command = SaveSetting

Arguments: {#}

--Save Setting as full path filename--

Command = SaveSetting

Arguments: {full path filename}

Commands for analysis (Useful for batch processing):

--Open FLIM file--

Command = OpenFile

Arguments: {Full path to Filename}

Return: None

Note: Open .flim file and display

-- Read ImageJ format ROIs—

Command = ReadImageJROI

Arguments: {Filename}

Return: None

Note: Open ImageJ format ROI file (zip file)

-- Save ImageJ format ROIs—

Command = SaveImageJROI

-- Save ROIs—

Command = SaveROIs

-- Batch Processing—

Command = BatchProcessing

--Bin Pages/Frames—

Command = BinFrames

Arguments: {Number of frames to be binned}, {output filename} (optional)

Return: None

Note: Bin frames or pages to increase signal-to-noise ratio.

Same as “Bin time course...” menu.

--Calculate time course—

Command = CalcTimeCourse

Arguments: None

Note: Same as pressing button of “Calc

--Set FLIM intensity ranges—

Command = SetFLIMIntensityOffset

Arguments: Lower threshold, Higher threshold

Example: SetFLIMIntensityOffset, 20, 60

--Set Fit parameters--

Command = SetFitParams

Arguments: tau1, tau2, tauG, T0

Example: SetFitParams, 2.6, 1.1, 0.3, 1.0

--Fix Tau for fitting—

Command = FixTau

Arguments: Tau1, Tau2

Example: FixTau, 2.6, 1.1

--Set max values of intensity and lifetime (FLIM) --

Command = SetMaxFLIMIntensity

Arguments: {Int_thresh}, {LT_thresh}

--Set min values of intensity and lifetime (FLIM) --

Command = SetMinFLIMIntensity

Arguments: {Int_thresh}, {LT_thresh}

--Set fit range--

Command = SetFitRange

Arguments: StartPoint, EndPoint (in time point)

Example: SetFitRange, 5, 60

--Fix tau all--

Command = FixTauAll

Arguments: 0 or 1

Note: Fix or unfix all tau to the current values.

--Fit data--

Command = FitData

Arguments: none

Note: it will fit the fluorescence lifetime curve with current setting.

--Set Channel--

Command = SetChannel

Arguments: 1, 2, or 12

Note: Set display channel.

--Apply fit offset--

Command = ApplyFitOffset

Arguments: none

Note: It will fit the data and then adjust the offset of FLIM image display.

--Fit each frame--

Command = FitEachFrame

Arguments: 0 or 1 (for off or on)

Example: FitEachFrame, 1

Note: If turned on, it will fit fluorescence lifetime curve for each frame.

--Align frames--

Command = AlignFrames

Arguments: output filename

--Get number of pages of the opened file.--

Command = GetNPages

Return: number of pages of the opened file.

--Save current FLIM image--

Command = SaveCurrentImage

Arguments: output filename

--Export current intensity image in TIFF--

Command = ExportCurrentIntensityImageInTIFF

Arguments: save channel (optional. If no arguments, it save all channels)

--Concatenate images in the current directory--

Command=ConcatenateImages

Arguments: output filename

*Note For concatenation, you need to call "GetAnalysisStatus" to make sure that the process is done.

--Get Analysis Status--

Command=GetAnalysisStatus

Arguments: none

Reply: either "None", "ConcatenateFiles", etc – this is for handling of multiple files.

--Recover ROIs--

Command = RecoverROIs

Arguments: none

--Remove All ROIs--

Command = Remove All ROIs

Arguments: none

--Shift ROI's positions (1 xy set for All, >1 for each ROI)--

Command = ShiftAllROIs

Arguments: {x shift}, {y shift}[, {x2 shift}, {y2 shift}...]

--Get current value on Realtime plot--

Command = GetRealtimeValue

Arguments: none

Reply: {value}

--Analysis>Delete current page--

Command = DeleteCurrentPage

Arguments: none

--Analysis>Blank current page--

Command = BlankCurrentPage

Arguments: none

--Analysis>Extract pages...--

Command = ExtractPages

Arguments: {start#}[- {end#}]

--Translate images (1 set for all pages, >1 set for each)--

Command = TranslateFrames

Arguments: {x_shift}, {y_shift}[, {x2_shift}, {y2_shift}...]

--Get current page--

Command = GetCurrentPage

Arguments: none

Reply: {page#}[- {page_end#}]

--Set current page(s) with page# (and page_end#)--

Command = SetPages

Arguments: {page#}[, {page_end#}]

--Set "File#" TextBox--

Command = SetFileCounter

Arguments: {file#}

--"Reset plot" button on Time course --

Command = ResetTimeCourse

Arguments: none

--"Calculate current page" button on Time course—

Command = CalcCurrentPage

Arguments: none

Table of commands

(Commands in red color were added in Ver4.0.4 or later)

(Blue variables from ver4.0.42)

No	explanation	Command	arguments format	return
1	Analysis>Align Frames	AlignFrames	{output filename}	
2	"Apply offset" button	ApplyFitOffset	--	
3	Analysis>Time binning...	BinFrames	BinFrames, {#bin}, {output filename}	
4	"Calculate current page" on Time course window	CalcTimeCourse	--	
5	clear all uncaging rois	ClearUncagingLocation	--	
6	create new uncaging rois	CreateUncagingLocation	CreateUncagingLocation, {x_pixel}, {y_pixel} [, {x2_pixel}, {y2_pixel}...]	CreateUncagingLoc, {x_frac}, {y_frac} [, {x2_frac}, {y2_frac}, , ,]
7	disconnect PIPE	Disconnect	--	
8	"Fit" button	FitData	--	
9		FitEachFrame	FitEachFrame, {bool}	
10	set and fix Tau1, Tau2	FixTau	FixTau, {Tau1}, {Tau2}	
11	fix/unfix all taus	FixTauAll	FixTauAll, {bool: 0/1}	
12		GetCurrentChannel	--	currentChannel, {ch}
13	get current stage position	GetCurrentPosition	--	CurrentPosition, {x_um}, {y_um}, {z_um}
14	same as above	GetMotorPosition	--	CurrentPosition, {x_um}, {y_um}, {z_um}
15	correcting x,y,z-offsets	GetCurrentPosition_um	--	CurrentPosition_um, {x_um}, {y_um}, {z_um}
16		GetFileCounter	--	FileCounter, {File#}
17	@zoom = 1	GetFOVXY	--	FovXYum, {x_um}, {y_um}
18		GetFullFileName	--	fullFileName, {current_file_path}
19		GetIntensityFilePath	--	IntensityFilePath, {file_path}
20		GetNPages	--	Npages, {#page}
21		GetResolutionXY	--	ResolutionXY, {x_pixel}, {y_pixel}
22		GetScanMirrorXY_um	--	ScanMirrorXY_um, {um value of State.Acq.XOffset}, {um value of State.Acq.YOffset}
23		GetScanVoltageMultiplier	--	ScanVoltageMultiplier, {State.Acq.scanVoltageMultiplier[0]}, {State.Acq.scanVoltageMultiplier[1]}
24	"Voltage range X" and "Y" on Scan tab	GetScanVoltageRangeReference	--	ScanVoltageRangeReference, {State.Acq.XMaxVoltage}, {State.Acq.YMaxVoltage}
25		GetScanVoltageXY	--	ScanVoltageXY, {State.Acq.XOffset}, {State.Acq.YOffset}
26		IsDORunning	--	IsDORunning, {bool}
27		IsGrabbing	--	IsGrabbing, {bool}
28		IsUncaging	--	IsUncaging, {bool}
29	load "Setting-#.ini"	LoadSetting	LoadSetting, {#}	Setting, {#}
30	default path: Documents\FLIMage\Init_Files\	LoadSetting	LoadSetting, {file_name}	Setting, {file_name}
31	open .flim file	OpenFile	OpenFile, {full path filename}	
32	Convert a pixel pos to galvo voltages	PixelToVoltage	PixelToVoltage, {x_pixel}, {y_pixel}	PixelToVoltage, {x_volt}, {y_volt}
33		ReadImageROI	ReadImageROI, {ROI_file}	
34	stage control command	SetCenter	--	
35	set display channel	SetChannel	SetChannel, {1/2/12}	
36	export	SetChannelsToBeSaved	SetChannelsToBeSaved, {1/2/-1(all channels, default)}	ChannelsToBeSaved, {1/2/-1}
37	set "Fit start" and "Fit end"	SetFitRange	SetFitRange, {Fit_start}, {Fit_end}	
38	set thresholds of intenisty and lifetime (FLIM)	SetFLIMIntensityOffset	SetFLIMIntensityOffset, {min_thresh}, {max_thresh}	
39	not work? use "State.Acq.field_of_view = [x, y]"	SetFOVXY	SetFOVXY, {x_um}, {y_um}	FovXYum, {x_um}, {y_um}
40	export	SetIntensitySaving	SetIntensitySaving, {bool}	IntensitySaving, {bool}
41		SetMotorPosition	SetMotorPosition, {x_um}, {y_um}, {z_um}	SetMotorPositionDone, {x_um}, {y_um}, {z_um}

42	set x,y-scan pixels	SetResolutionXY	SetResolutionXY, {x_pixel}, {y_pixel}	ResolutionXY, {x_pixel}, {y_pixel}
43		SetScanMirrorXY_um	SetScanMirrorXY_um, {x_um}, {y_um}	ScanVoltageXY, {State.Acq.XOffset}, {State.Acq.YOffset}
44		SetScanVoltageXY	SetScanVoltageXY, {x_volt}, {y_volt}	ScanVoltageXY, {State.Acq.XOffset}, {State.Acq.YOffset}
45	set current uncaging position	SetUncagingLocation	SetUncagingLocation, {x_pixel}, {y_pixel}	UncagingLocation, {x_pixel}, {y_pixel}
46		SetZoom	SetZoom, {value}	Zoom, {value}
47	set Z-Stack and #Slices	SetZSliceNum	SetZSliceNum, {#ZSlice}	ZSliceNum, {#ZSlice}
48		StartDO	--	
49		StartGrab	--	
50		StartLoop	--	
51	Start button on Uncaging Control Panel	StartUncaging	--	
52		StopDO	--	
53		StopGrab	--	
54		StopLoop	--	
55		StopUncaging	--	
56	checkbox of "Uncage while imaging"	Uncaging	Uncaging, {bool:0/1}	
62		AcquisitionDone	--	ParameterFileSaved, {file_name}
63		DODone	--	DODone
64		ExtCommandExecuted	--	ExtCommandExecuted
65		FLIMImageStarted	--	FLIMImageStarted
66		FocusStart	--	FocusStart
67		FocusStop	--	FocusStop
68		FrameAcquisitionDone	--	FrameAcquisitionDone, {current_frame}, {total #Slices}
69		GrabAbort	--	GrabAbort
70		GrabStart	--	GrabStart
71		ParametersChanged	--	ParameterFileSaved, {file_path}
72		ReadFileDone	--	ParameterFileSaved, {file_name}
73		SaveImageDone	--	SaveImageDone
74		SliceAcquisitionDone	--	SliceAcquisitionDone, {current_slice}, {total #Slices}
75		SliceAcquisitionStart	--	SliceAcquisitionStart
76		StageMoveDone	--	StageMoveDone, {x_um}, {y_um}, {z_um}
77		StageMoveStart	--	StageMoveStart
78		UncagingDone	--	UncagingDone
79		GetVersion	--	Version, {ver#}
80	set Fit parameters (tau1, tau2, tauG, T0)	SetFitParams	SetFitParams, {tau1}, {tau2}, {tauG}, {T0}	
81	Checkbox of "Hold This Image"	HoldThisImage	HoldThisImage, {bool: 0/1}	
82	set max values of intenisty and lifetime (FLIM)	SetMaxFLIMIntensity	SetMaxFLIMIntensity, {Int_thresh}, {LT_thresh}	
83	set min values of intenisty and lifetime (FLIM)	SetMinFLIMIntensity	SetMinFLIMIntensity, {Int_thresh}, {LT_thresh}	
84	Set DIO output on DIO panel	SetDIOPanel	SetDIOPanel, {port#:0-6}, {bool:0/1} [, {port#:0-6}, {bool:0/1}...]	
85	Save current FLIM image	SaveCurrentImage	SaveCurrentImage, {output filename}	
86	Concatenate images in the current directory	ConcatenateImages	ConcatenateImages, {output filename}	
87	Move Piezo motor one step	MovePiezoStep	MovePiezoStep, {float: micrometer}	
88	Export current intensity image in TIFF	ExportCurrentIntensityImageInTIFF	ExportCurrentIntensityImageInTIFF[, {ch#}]	
89	Get Analysis Status	GetAnalysisStatus	--	"None"/"ConcatenateFiles"/etc
90	Save ROI as ImageI	SaveImageROI	--	
91	Save ROIs	SaveROIs	--	
92	Batch processing (Analysis)	BatchProcessing	--	
93	Recover ROIs	RecoverROIs	--	
94	Recover ROIs	RemoveAllROIs	--	

95	Shift ROI's positions (1 xy set for All, >1 for each ROI)	ShiftAllROIs	ShiftAllROIs, {x shift}, {y shift}[, {x2 shift}, {y2 shift}...]	
96	Get current value on Realtime plot	GetRealtimeValue	--	GetRealtimeValue, {value}
97	"Focus" button	Focus	--	
98	"Stop" button in Stage panel	StopMotor	--	
99	"Zero all" button in Stage panel	SetZeroAll	--	
100	"Analyze after each acquisition" checkbox	SetAnalyzeAfterAcq	SetAnalyzeAfterAcq, {bool: 0/1}	
101	Save "Setting-#.ini"	SaveSetting	SaveSetting, {#}	
102	Save Setting as full path filename	SaveSetting	SaveSetting, {full path filename}	
103	Analysis>Delete current page	DeleteCurrentPage	--	
104	Analysis>Blank current page	BlankCurrentPage	--	
105	Analysis>Extract pages...	ExtractPages	ExtractPages, {start#}[- {end#}]	
106	Translate images (1 set for all pages, >1 set for each)	TranslateFrames	TranslateFrames, {x_shift}, {y_shift}[, {x2_shift}, {y2_shift}...]	
107		GetCurrentPage	--	CurrentPage, {page#}[- {page_end#}]
108	Set current page(s) with page# (and page_end#)	SetPages	SetPages, {page#}[, {page_end#}]	
109	Set "File#" TextBox	SetFileCounter	SetFileCounter, {file#}	
110	"Reset plot" button on Time course	ResetTimeCourse	--	
111	"Calculate current page" button on Time course	CalcCurrentPage	--	

Using python example code for PIPE

You can run attached FLIM_pipeClient.py class.

Once you run it you can:

```
exec(open("FLIM_pipeClient.py").read())
value = flim.sendCommand('State.Acq.msPerLine?') #Return '2'
flim.sendCommand('State.Acq.msPerLine = 2') #Change msPerline to 2
ms.
flim.sendCommand('StartGrab') #Will grab.
flim.sendCommand('SetMotorPosition,12,89.2,0') #set motor position.
```